

# SLACK\_SETUP

## Slack Integration Setup Guide

This guide will walk you through setting up Slack notifications for HelixCode.

### Prerequisites

- Slack workspace admin access
- Ability to create incoming webhooks

### Step 1: Create Slack App

1. Go to <https://api.slack.com/apps>
2. Click **“Create New App”**
3. Select **“From scratch”**
4. Enter app name: HelixCode Notifications
5. Select your workspace
6. Click **“Create App”**

### Step 2: Enable Incoming Webhooks

1. In your app settings, navigate to **“Incoming Webhooks”** (left sidebar)
2. Toggle **“Activate Incoming Webhooks”** to **On**
3. Scroll down and click **“Add New Webhook to Workspace”**
4. Select the channel where you want notifications (e.g., #helix-notifications)
5. Click **“Allow”**

### Step 3: Copy Webhook URL

1. You'll see your webhook URL listed (it looks like: `https://hooks.slack.com/services/T000000000/B000000000/XXXXXXXXXXXXX`)
2. Click **“Copy”** to copy the webhook URL

### Step 4: Configure HelixCode

#### Option A: Using Environment Variables

Add to your `.env` file:

```
HELIX_SLACK_WEBHOOK_URL=https://hooks.slack.com/services/YOUR/  
WEBHOOK/URL
```

## Option B: Using config.yaml

Edit config/config.yaml:

```
notifications:  
  enabled: true  
  channels:  
    slack:  
      enabled: true  
      webhook_url: "${HELIX_SLACK_WEBHOOK_URL}" # Or paste URL  
        directly (not recommended for security)  
      channel: "#helix-notifications"  
      username: "HelixCode Bot"  
      timeout: 10
```

## Step 5: Test Integration

### Using HelixCode CLI (if available)

```
helix notify test --channel slack --message "Test notification"
```

### Using API

```
curl -X POST http://localhost:8080/api/v1/notifications/test \  
  -H "Content-Type: application/json" \  
  -H "Authorization: Bearer YOUR_TOKEN" \  
  -d '{  
    "channel": "slack",  
    "title": "Test Notification",  
    "message": "Testing Slack integration from HelixCode",  
    "type": "info"  
  }'
```

### Using Go Code

```
import "dev.helix.code/internal/notification"  
  
// Create notification engine  
engine := notification.NewNotificationEngine()  
  
// Register Slack channel  
slackChannel := notification.NewSlackChannel(  
  engine,  
  "#helix-notifications",  
  "HelixCode Bot",  
  10,  
)
```

```

        "https://hooks.slack.com/services/...",
        "#helix-notifications",
        "HelixCode Bot",
    )
    engine.RegisterChannel(slackChannel)

    // Send test notification
    testNotif := &notification.Notification{
        Title:    "Test Notification",
        Message:  "Testing Slack integration",
        Type:     notification.NotificationTypeInfo,
    }
    err := engine.SendDirect(context.Background(), testNotif,
        []string{"slack"})

```

## Step 6: Configure Notification Rules

Edit `config.yaml` to define when to send Slack notifications:

```

notifications:
  rules:
    - name: "Critical Failures to Slack"
      condition: "type==error"
      channels: ["slack"]
      priority: urgent
      enabled: true

    - name: "Success Notifications"
      condition: "type==success"
      channels: ["slack"]
      priority: medium
      enabled: true

```

## Notification Types & Icons

HelixCode automatically selects appropriate icons for each notification type:

| Type    | Icon                                                                                                     | Example Use Case            |
|---------|----------------------------------------------------------------------------------------------------------|-----------------------------|
| info    |  :information_source: | General information         |
| success | :white_check_mark:                                                                                       | Task completed successfully |
| warning | :warning:                                                                                                | Non-critical issues         |
| error   | :x:                                                                                                      | Task failures, errors       |

| Type  | Icon             | Example Use Case       |
|-------|------------------|------------------------|
| alert | :rotating_light: | Critical system alerts |

## Troubleshooting

### Error: “invalid\_token”

- **Cause:** Webhook URL is incorrect or invalid
- **Solution:** Verify the webhook URL is copied correctly from Slack

### Error: “channel\_not\_found”

- **Cause:** The specified channel doesn’t exist or bot lacks access
- **Solution:** Verify the channel exists and the webhook is configured for it

### No notifications appearing

- **Cause:** Channel might be disabled in config
- **Solution:** Check `enabled: true` in `config.yaml`

### Rate limiting issues

- **Cause:** Slack has a rate limit of 1 message per second per webhook
- **Solution:** HelixCode automatically handles rate limiting (future feature)

## Advanced Configuration

### Custom Username & Icon

```
slack:
  username: "My Custom Bot"
  # Note: icon_emoji is set automatically based on notification
  type
```

### Multiple Slack Channels

You can create multiple Slack channels for different purposes:

```
// Production alerts
prodChannel := notification.NewSlackChannel(
  webhookProd,
  "#production-alerts",
  "Production Bot",
```

```
)  
  
// Development notifications  
devChannel := notification.NewSlackChannel(  
    webhookDev,  
    "#dev-notifications",  
    "Dev Bot",  
)
```

## Future: Slack Block Kit (Rich Formatting)

Future versions will support Slack's Block Kit for rich, interactive messages with: -  
Structured layouts - Action buttons - Context blocks - File attachments

## Security Best Practices

- 1. Never commit webhook URLs to version control**
  - Use environment variables
  - Add .env to .gitignore
- 2. Rotate webhook URLs regularly**
  - Delete old webhooks in Slack
  - Generate new ones periodically
- 3. Use separate webhooks for different environments**
  - Production webhook → #production-alerts
  - Staging webhook → #staging-alerts
  - Development webhook → #dev-notifications

## Resources

- [Slack Incoming Webhooks Documentation](#)
- [Slack Block Kit Builder](#)
- [HelixCode Notification API Documentation](#)

## Support

If you encounter issues: 1. Check the [Troubleshooting](#) section 2. Review HelixCode logs for error messages 3. Open an issue on GitHub